Final Project Report

STOR 323: Ethics of Machine Learning in Society

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1 Introduction

Research Question: **Can mathematically calibrated loss functions reduce overconfidence in language models, thus mitigating (the ethical harms of) cognitive offloading?"**

As Artificial Intelligence (AI) advances, a central ethical concern is **cognitive offloading**—humans transferring thought processes to external tools and thus placing **blind trust in AI**. Such trust becomes risky when models **hallucinate** [5], confidently producing answers without adequate knowledge.

Studies show that younger users display "higher AI tool usage and cognitive offloading, but lower critical thinking scores" [2]. While AI enhances efficiency, this reliance can reduce autonomy and promote uncritical acceptance of outputs. At the same time, AI can *"enhance human creativity through higher-level co-creation"* [1], underscoring the need for balance between support and dependence.

To preserve human agency, we argue for rethinking how models are trained. This can be through modified loss functions or alternative training data that encourages engagement rather than passive trust. The goal is to balance capability with accountability so that performance advances without compromising ethical alignment.

2 Ethics Overview

There are two primary ethical concerns associated with offloading and the potential spread of misinformation that comes with it. Each can be viewed from the perspective of deontology, consequentialism, and virtue ethics while offering valuable insight into the ethical need of reshaping use of AI.

1. **Diminished User Autonomy**
   From a **deontological view**, letting AI "think for us" (especially when information is wrong) undermines our duty to act as rational, autonomous agents, turning humans into passive followers of algorithms. Using AI may also be viewed as a means to an end, using it to get what you want by taking shortcuts, instead of doing what's right. **A consequentialist** would emphasize the social harms of this dependence, such as manipulation, loss of agency, and spread of misinformation outweigh any short-term benefits. **Virtue ethics** further sees autonomy as a moral excellence rooted in prudence and self-control, both of which are diminished by intellectual passivity. Together these perspectives argue that ethical AI use must preserve human

agency, encouraging people to think with AI in order to sustain autonomy, responsibility, and virtuous self-direction.

2. **Educational inadequacies**
   A **deontologist views** educators as having a duty to foster students' rational and moral growth, and uncritical AI use violates this duty by reducing learners to passive recipients rather than autonomous thinkers. A **consequentialist argues** that long-term harms—such as reduced creativity, critical thinking, and learning inaccurate concepts—outweigh the short-term gains of efficiency. **Virtue ethics** adds that education should cultivate intellectual virtues like curiosity, perseverance, and honesty, all of which are weakened when students rely on AI to think for them. Similarly to diminished user autonomy, these perspectives emphasize that AI in education should enhance wisdom and autonomy, serving as a partner in human learning rather than a substitute for it.

3 Literature Survey

3.1 Background on Training Paradigms

There exist several popular methods to drive reinforcement learning in machine/deep learning algorithms. We aim to establish a basis for understanding based on the information presented in our literature review.

One of the most widely used loss functions for training deep language models is the *Cross-Entropy (CE) Loss*, which corresponds to the objective of *Maximum Likelihood Estimation (MLE)* for next-token prediction. In early generations of large language models (LLMs), this loss was optimized to minimize the expected value between the predicted distribution and the actual data distribution.

Formally, given a sequence of tokens $x = (x_1, x_2, \ldots, x_T)$, the MLE objective is defined as:

$$\mathcal{L}_{\text{MLE}}(\theta) = E_{x \sim p_{\text{data}}} \left[-\sum_{t=1}^{T} \log p_\theta(x_t \mid x_{<t}) \right] \tag{1}$$

When trained purely under (1), without explicit regularization or preference-based feedback, models can become *overconfident in incorrect predictions*; a common issue observed in early GPT-type architectures. With regard to Cognitive Offloading, this can create an ethical concern where, if users are blindly trusting the outputs of these models, they are in essence spreading misinformation.

Later model families introduced explicit regularization using *Kullback–Leibler (KL) divergence*, denoted via:

$$D_{\text{KL}}(P \,\|\, Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)} \tag{2}$$

On its own, the cross-entropy loss function implicitly performs KL divergence minimization. However, because this optimization occurs over the empirical data distribution, it tends to produce overconfident predictions, highlighting the need for additional regularization mechanisms in order to combat the ethical concerns surrounding cognitive offloading.

During subsequent reinforcement learning and preference alignment phases (e.g., via Reinforcement Learning from Human Feedback, RLHF), the relationship between CE loss and KL divergence is

made explicit by including a KL regularizer as a part of a new objective:

$$\mathcal{L}_{\text{RLHF}} = -E_{x\sim\pi_{\text{new}}}\big[r(x)\big] + \beta D_{\text{KL}}(\pi_{\text{new}} \,\|\, \pi_{\text{old}}) \tag{3}$$

While reinforcement learning from human feedback (RLHF) has proven highly effective as an alignment mechanism, it remains largely an engineering construct—a practical intervention rather than a fully principled mathematical solution. Conceptually, RLHF mitigates cognitive offloading behaviors by embedding human evaluative priors into the optimization process, yet it does not directly resolve the underlying issues of uncertainty calibration or internal reasoning consistency.

In summary, this progression from maximum likelihood estimation to reinforcement learning-based alignment methods embodies efforts toward mitigating certain risks of overly confident behavior in language models. While (1) is good in capturing linguistic structure , it tends to amplify issues of cognitive offloading by allowing humans to uncritically adopt the outputs of AI without verifying either their factual or contextual validity. (3) introduces preference-guided fine-tuning into the mix to balance out such overconfidence, embedding human evaluative signals into models that encourage better-calibrated reasoning. Still the challenge remains to find a mathematical solution to combat these issues.

3.2 Linearly Adaptive Cross-Entropy Loss

The standard cross-entropy (CE) loss encourages the model to match the empirical target distribution by maximizing the likelihood of the correct class:

$$\mathcal{L}_{\text{CE}}(P, Q) = -\sum_i P(x_i) \log Q(x_i), \tag{4}$$

where P is the true label distribution and Q represents the predicted probabilities. While effective in many settings, this approach treats all samples uniformly, regardless of how confidently the model already identifies the correct class. Consequently, the network can produce excessively peaked outputs, particularly during later stages of training, where easy examples continue to dominate the gradient. This behavior often results in overconfident predictions and poorly calibrated probabilities.

To address these issues, [3] introduces a linearly adaptive variant of cross-entropy, known as LACE, a confidence-dependent scaling function that modulates the contribution of each sample. Let $Q(x_c)$ denote the predicted probability of the correct class x_c. The LACE loss can be written as:

$$\mathcal{L}_{\text{LACE}}(P, Q) = -\left[1 - Q(x_c)\right] \log Q(x_c). \tag{5}$$

In this formulation, although the loss function is expressed as $\mathcal{L}_{\text{LACE}}(P, Q)$, the optimization is effectively carried out only with respect to the model parameters θ governing the predicted distribution Q_θ. The true label distribution P serves only as a fixed, one-hot reference indicating the correct class index x_c. Consequently, the gradient of the loss depends solely on $Q_\theta(x_c)$, meaning that the learning process minimizes

$$\min_\theta \; \mathcal{L}_{\text{LACE}}(Q_\theta) = \min_\theta \; -\big[1 - Q_\theta(x_c)\big] \log Q_\theta(x_c).$$

This formulation preserves the notational consistency of traditional cross-entropy while clarifying that the model is not optimized over both P and Q, but rather only over the predicted distribution Q_θ. The inclusion of the $(1 - Q_\theta(x_c))$ term further introduces a confidence-dependent scaling effect, reducing the gradient magnitude for well-classified samples and increasing it for uncertain or misclassified ones, thereby promoting a more stable and balanced learning process.

In practice, the linear scaling in LACE addresses several limitations of standard cross-entropy:

- **Stability of Confidence Trajectories**
  The diminishing scale factor reduces the runaway sharpening effect of standard CE, producing smoother probability dynamics and improving calibration at high confidence levels.

- **Reweighting Ambiguous Samples**
  Examples with lower predicted certainty receive greater emphasis, ensuring that hard or ambiguous cases retain influence throughout optimization rather than being overshadowed by easy samples early in training.

- **Mitigation of Overconfident Failures**
  By discouraging extremely peaked predictions, the model is less prone to brittle or biased outputs. This implicit regularization enhances robustness under mild distribution shifts without requiring auxiliary temperature-based corrections.

Conceptually, LACE behaves like a self-regulating likelihood term: the model gauges its own certainty before deciding how strongly to adjust parameters. Gradients are softened when predictions are confident and amplified when uncertainty exists, preserving the computational simplicity of cross-entropy while reducing the pathological incentives that lead to unwarranted certainty.

3.3 Generalized Kullback–Leibler (GKL)

As noted earlier, one of the primary pitfalls of (1) on its own is overconfidence in predictions, trying to perfectly match the empirical data distribution. The addition of KL divergence to MLE extends this to soft targets, with both ultimately suffering from optimization asymmetry, entropy imbalance, and sample-wise bias. In practice, this means that while models trained under these objectives may converge more smoothly, they often produce poorly calibrated probabilities, especially in regions of sparse data.

A high-level, reinforcement learning algorithm we observed in [4] that combats these issues is *Generalized Kullback–Leibler (GKL)* modeled via:

$$\mathcal{L}_{\mathrm{GKL}}(x_m, x_n) = \frac{\alpha}{4} \left\| \sqrt{\varphi^*(x_m, x_n)}\, (\Delta\mathbf{m} - \Delta\mathbf{n}) \right\|^2 - \beta\, \mathcal{S}(\mathbf{s}_m^\top) \log \mathbf{s}_n \tag{6}$$

The authors of this paper propose a new loss function, rooted in principles discussed in previous sections. The main idea behind is to take the standard Cross-Entropy loss (denoted by $\mathcal{S}(\mathbf{s}_m^\top) \log \mathbf{s}_n$) and scale it by some hyperparameter β while extending the standard KL divergence with weighted MSE. This plays a regularizing role, balancing fidelity with a weighted MSE term that aligns local logits. In depth (6):

- **Breaks Asymmetric Optimization**

  - Introduces symmetric gradient flow, allowing both sides of the comparison (teacher and student models) to be co-optimized.
  - This symmetry restores balance between local structural alignment (MSE term) and global probabilistic alignment (Cross-Entropy term).

- **Introduces a Smoother, Class-aware Weight Function**
  - The weight function

$$\varphi^*(x_m, x_n) = \begin{cases} \sqrt{\mathcal{S}(\mathbf{w}_m)^\gamma}, & \text{(sample-wise)} \\ \sqrt{\mathcal{S}(\overline{\mathbf{w}}_m)^\gamma}, & \text{(class-wise)} \end{cases}$$

    injects *global, class-level information* into the optimization which, in turn, reduces over-penalization of confident predictions and mitigates sample bias

- **Balances Local and Global Learning**
  - Local consistency (via weighted MSE): aligns fine-grained logit relationships between pairs of classes.
  - Global consistency (via Cross-Entropy): preserves overall probability distribution structure.

Intuitively, (6) teaches models how to be confidently uncertain by tempering overconfidence from CE, balancing asymmetric optimization from KL, and regularizing representations across the feature space through structured weighting.

4 Use case of each loss function

We aim to provide a small scale implementation of each loss function to test their effectiveness in measuring uncertainty and therefore, mitigating some of the ethical concerns surrounding cognitive offloading.

Each loss function was trained with the same set of in-domain questions; these consisted of medical questions (e.g., "What are the symptoms of the flu?"). The major point of the tests were to see each models confidence (hopefully lack thereof) on out-of-domain questions the models have never seen. We used a GPT-2 architecture with 4 layers and 256 embedding dimensions, trained for 20 epochs with AdamW optimization (learning rate: 5×10^{-4}). To combat random seeding, each loss function had 3 trials averaged to measure overall performance. Evaluation metrics were normalized combined uncertainty (Epistemic + aleatoric).

4.1 Lace

LACE alone was insufficient for uncertainty recognition, as the model would default to its pre-trained knowledge when faced with out-of-domain queries, generating confident but potentially incorrect responses. To address this limitation, we had to combine LACE with two key engineering techniques:

1. **Monte Carlo Dropout** for inference-time uncertainty estimation, performing 15 forward passes with dropout enabled ($p = 0.15$) to measure epistemic uncertainty through prediction variance and response diversity.

2. **No explicit uncertainty training**: Unlike typical approaches that train models to respond "I don't know" on out-of-domain examples, our model was trained *exclusively* on medical data, demonstrating emergent uncertainty detection on never-seen domains purely through MC Dropout variance.

Table 1: Uncertainty Quantification: Standard CE vs LACE (3 Trial Average)

Question	**CE Unc.**	**LACE Unc.**	**Status**
In-Domain (Medical - Trained)			
What is diabetes?	0.105	0.152	Confident
What are symptoms of flu?	0.317	0.380	Uncertain
What is pneumonia?	0.068	0.174	Confident
How to prevent heart disease?	0.036	0.116	Confident
What is arthritis?	0.019	0.128	Confident
Out-of-Domain (Physics/CS - Never Seen)			
What is quantum entanglement?	0.353	**0.395**	**Uncertain**
How does blockchain work?	0.084	**0.286**	Confident
What is machine learning?	0.201	**0.302**	Confident
Explain general relativity?	0.363	**0.439**	**Uncertain**
What is photosynthesis?	0.244	**0.329**	Confident
In-Domain Avg	0.109	0.190	–
OOD Avg	0.249	**0.350**	–

Threshold for uncertainty: 0.35. **Key finding**: LACE produces higher absolute uncertainty on OOD questions (0.350 vs CE's 0.249). LACE correctly flags 2/5 OOD questions as uncertain.

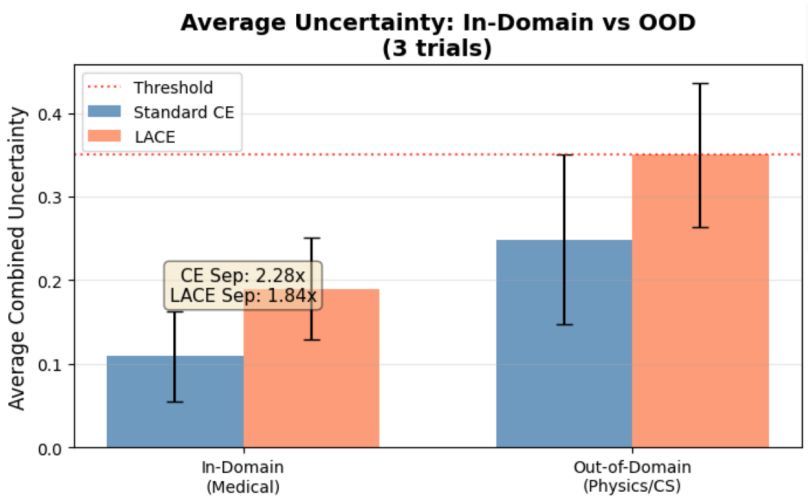

Figure 1: Average Uncertainty: LACE vs. Standard Cross Entropy

Overall, our experiments revealed that LACE is seemingly most effective when used within a **multi-component training framework**, consistent with reasoning models such as OpenAI's o1-preview and DeepSeek-R1, which combine specialized loss functions with inference-time computation and reinforcement learning from human feedback.

We recognize limitations in our implementation, such as small training data and approximate hyperparameters, but an overall trend in higher uncertainty shows promise for LACE's ability to combat ethical concerns of cognitive offloading.

4.2 Generalized Kullback–Leibler (GKL) Loss Results

Similar to our testing of LACE, we used the same questions with the same ratio and implemented MC Dropout for reasoning purposes. We were however not as successful.

Table 2: Uncertainty Quantification: Standard CE vs GKL (3 Trial Average)

Question	**CE Unc.**	**GKL Unc.**	**Status**
In-Domain (Medical - Trained)			
What is diabetes?	0.002	0.589	Over-uncertain
What are symptoms of flu?	0.111	0.920	Over-uncertain
What is pneumonia?	0.002	0.596	Over-uncertain
How to prevent heart disease?	0.002	0.589	Over-uncertain
What is arthritis?	0.003	0.599	Over-uncertain
Out-of-Domain (Physics/CS - Never Seen)			
What is quantum entanglement?	0.183	0.981	Over-uncertain
How does blockchain work?	0.025	0.871	Over-uncertain
What is machine learning?	0.111	0.974	Over-uncertain
Explain general relativity?	0.162	0.974	Over-uncertain
What is photosynthesis?	0.110	0.973	Over-uncertain
In-Domain Avg	0.024	0.659	–
OOD Avg	0.118	0.955	–
Separation	**4.98x**	1.45x	–

While GKL shows higher absolute OOD uncertainty (0.955 vs 0.118), it also exhibits excessive in-domain uncertainty (0.659 vs 0.024), resulting in poor separation (1.45x vs CE's 4.98x). Standard CE achieves better relative discrimination between in-domain and OOD queries.

Our experiments with the Generalized Kullback–Leibler (GKL) loss reveal a critical limitation when applied to uncertainty quantification: while GKL successfully produces higher uncertainty on out-of-domain queries compared to in-domain medical questions (normalized entropy: 0.955 vs 0.659), the entropy regularization component causes excessive global uncertainty, with models approaching near-uniform distributions (0.59–0.98 out of maximum 1.0). This results in poor calibration for threshold-based detection (separation factor: 1.45x), suggesting that loss functions designed for knowledge distillation require substantial modification when repurposed for uncertainty awareness.

5 Impact on Cognitive Offloading

The transition from basic Maximum Likelihood Estimation (MLE) to advanced regularization techniques like Linearly Adaptive Cross-Entropy (LACE) and Generalized Kullback–Leibler (GKL) directly addresses the core technical enabler of harmful cognitive offloading: the machine's unwarranted confidence [9, 5].

5.1 The Role of Advanced Loss Functions

The Linearly Adaptive Cross-Entropy (LACE) loss function adjusts sample weighting based on the model's confidence, which helps stabilize predictions and improve calibration. This approach reduces overconfidence and false certainty, promoting responsible communication of uncertainty. Similarly, the Generalized Kullback–Leibler (GKL) loss function balances cross-entropy and KL divergence terms through structured weighting to regularize representations and moderate overconfidence. Although we were unsuccessful in our implementation, it's theoretical promise should continue to be examined.

5.2 Preserving Human Agency

By making AI models more transparent about their own uncertainty, these modified loss functions indirectly support the ethical battle to preserve human agency and autonomy. When models are less overconfident, users are less likely to fall into the trap of "intellectual passivity" and are instead encouraged to think with AI rather than letting it think for them [6, 7]. This technical shift helps to transform AI from a potential substitute for critical thought into a partner in human learning [2].

6 Future work

6.1 Expanded Empirical Evaluation

The analysis in this report is conceptual and literature based in addition to a very small scale implementation. Next-steps are to verify the conclusions made empirically and scale the testing the proposed loss functions (LACE, GKL) on real world LLMs. This matters because the actual change of calibration, hallucination rates, or model behavior is the way to definitely prove that these loss functions actually reduce cognitive offloading. There are several phenomena in learning, as well as a lack of vast training data, that serves as a limitation to our survey. Nonetheless, we believe this is a foundation for future work.

6.2 Human-AI Interaction Studies

The empirical evidence proposed in the previous section is proof that the models change, but these technical calibration improvements only matter if they change human behavior. This is why we think that part of the next-steps in research should be analysis of how users respond to properly calibrated AI versus overconfident AI. A better understanding of this can impact how we move forward with testing and creating models. Since cognitive offloading is a behavioral phenomenon, work must be done cross-disciplinary with groups such as neuro-scientists to measure effectiveness.

7 References

[1] Haase, J. and Pokutta, S. (2024). *Human-AI Co-Creativity: Exploring Synergies Across Levels of Creative Collaboration.* arXiv preprint. Available at: https://arxiv.org/html/2411.12527v2.

[2] Gerlich, M. (2025). *AI Tools in Society: Impacts on Cognitive Offloading and the Future of Critical Thinking. Societies*, 15(1), 6. Available at: https://www.mdpi.com/2075-4698/15/1/6.

[3] Shim, J.W. (2024). *Enhancing Cross Entropy with a Linearly Adaptive Loss Function for Optimized Classification Performance. Scientific Reports*, 14, 27405. Available at: https://www.nature.com/articles/s41598-024-78858-6.

[4] Cui, J., Zhu, B., Xu, Q., Tian, Z., Qi, X., Yu, B., Zhang, H., and Hong, R. (2025). *Generalized Kullback–Leibler Divergence Loss.* arXiv preprint. Available at: https://arxiv.org/abs/2503.08038.

[5] Kalai, A. T., Nachum, O., Vempala, S. S., and Zhang, E. (2025). *Why Language Models Hallucinate.* arXiv preprint. Available at: https://arxiv.org/abs/2509.04664.

[6] Risko, E. F. and Gilbert, S. J. (2016). *Cognitive Offloading. Trends in Cognitive Sciences*, 20(9), 676–688. Available at: https://doi.org/10.1016/j.tics.2016.07.002.

[7] Clark, A. and Chalmers, D. (1998). *The Extended Mind. Analysis*, 58(1), 7–19. Available at: https://doi.org/10.1093/analys/58.1.7.

[8] Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C. L., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Askell, A., Welinder, P., Christiano, P., Leike, J., and Lowe, R. (2022). *Training Language Models to Follow Instructions with Human Feedback.* arXiv preprint. Available at: https://arxiv.org/abs/2203.02155.

[9] Desai, S. and Durrett, G. (2020). *Calibration of Pre-trained Transformers. Findings of EMNLP.* Available at: https://arxiv.org/abs/2003.07892.

[10] Kumar, A., Raghunathan, A., Jones, R., Ma, T., and Liang, P. (2023). *Fine-Tuning Can Distort Pretrained Features and Underperform Out-of-Distribution. NeurIPS.* Available at: https://arxiv.org/abs/2202.10054.

[11] Raji, I. D. and Buolamwini, J. (2019). *Actionable Auditing: Investigating the Impact of Publicly Naming Biased Performance Results. AAAI/ACM Conference on AI Ethics.* Available at: https://doi.org/10.1145/3306618.3314244.

[12] Floridi, L. and Cowls, J. (2022). *A Unified Framework of Five Principles for Ethical AI. Harvard Data Science Review.* Available at: https://doi.org/10.1162/99608f92.1f1f8810.